

Christof Teuscher

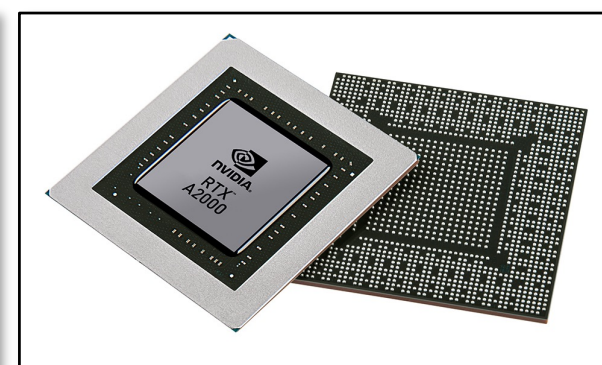
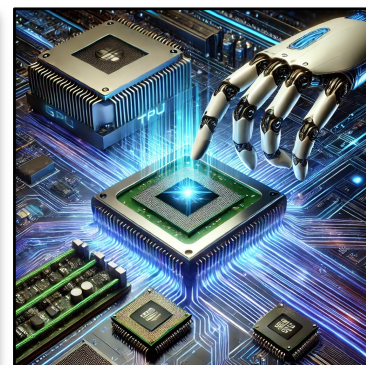
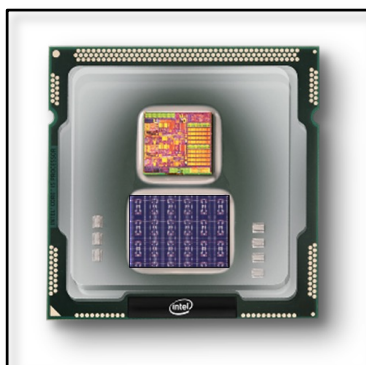
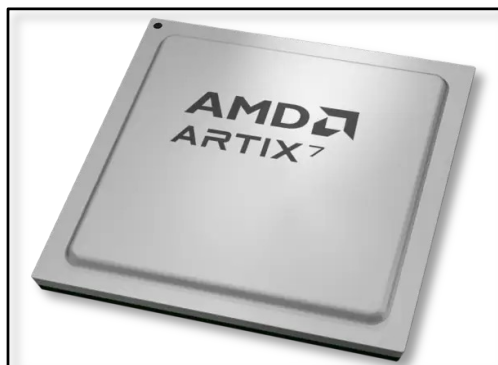
ECE 410/510: Hardware for AI and ML, Spring 2026

Week 7: Recap, outlook, and reminders

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What did you learn last week?

Recap

- How can we accelerate an algorithm?
- What are the most promising ways to do so today?
- What are the biggest bottlenecks in today's traditional computing architectures?
- What is a non-von Neumann architecture?
- What is in-memory computation.
- What is one way to realize in-memory computation?

Apple in Talks with Intel, Samsung for US Chipmaking

CIOTechOutlook Team | Tuesday, 05 May 2026, 18:28 PM IST



- Apple explores US chip partnerships with Intel Samsung
- Strategy aims to diversify semiconductor supply chains
- Talks support domestic manufacturing and resilience goals

Apple is reportedly in discussions with leading **semiconductor companies Intel and Samsung** to strengthen **chip manufacturing operations** in the United States, reflecting the company's broader strategy to diversify its **supply chain** and reduce dependence on overseas production.

The move comes amid increasing geopolitical tensions, supply chain

vulnerabilities, and growing pressure from the **US government** to boost **domestic semiconductor manufacturing**. By exploring partnerships with major **chipmakers, Apple** aims to secure more resilient production capabilities for its future devices while aligning with national efforts to enhance technological self-sufficiency.

Intel is considered a key contender in these discussions due to its expanding foundry **business and significant investments** in advanced **chip fabrication** facilities across the US. The company has been actively positioning itself as a **manufacturing partner** for **global technology firms** seeking localised production alternatives. A collaboration with Apple could provide Intel with a high-profile client while accelerating its ambitions in the foundry segment.

Samsung, another potential partner, already maintains a strong semiconductor presence in the United States and has announced substantial investments to expand its **manufacturing footprint**. Its expertise in advanced **chip production** and established relationship with Apple make it a strategic candidate for future collaboration.

Inference vs training in systolic array

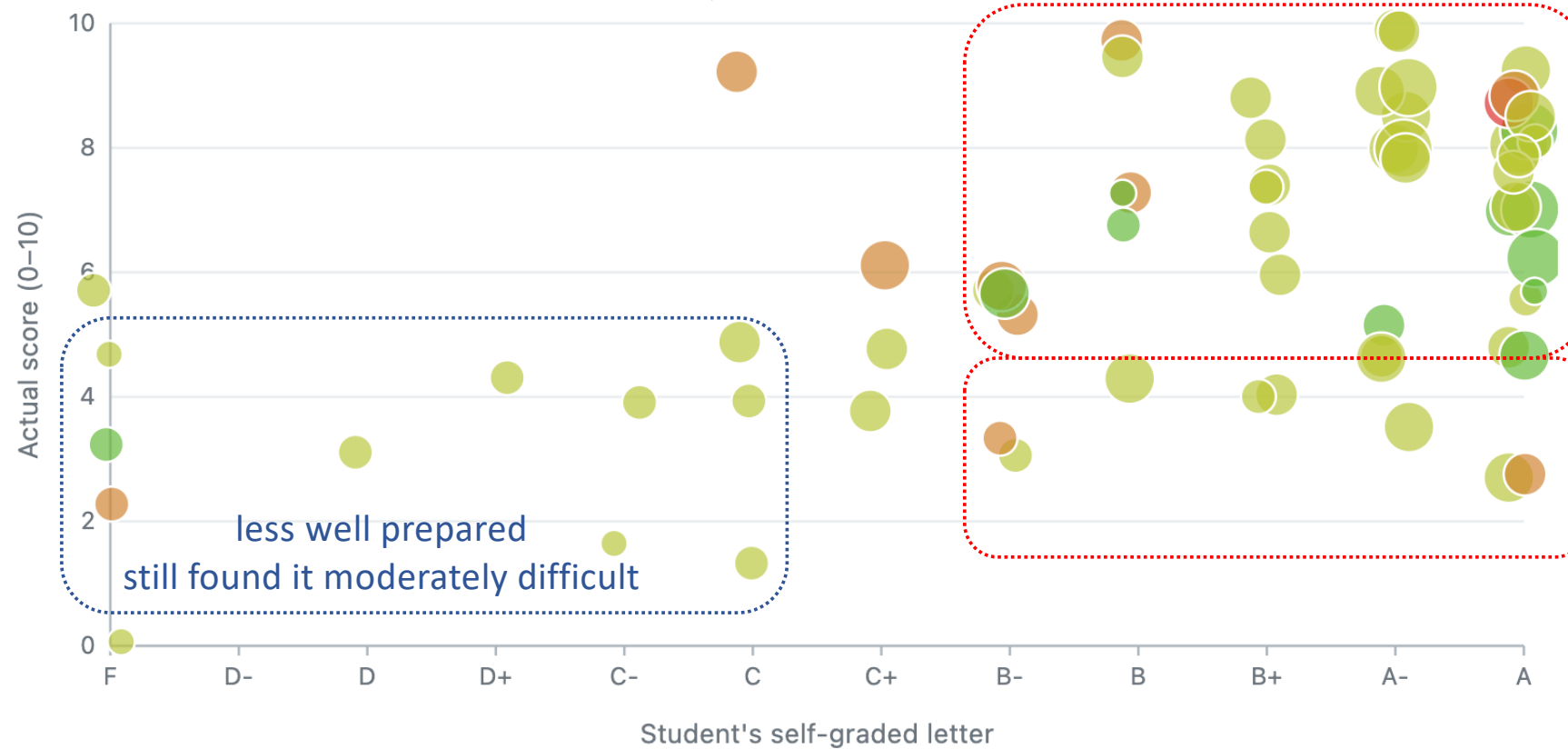
Inference

- **Weight-stationary dataflow**
 - Each PE holds a weight; activations flow through
- **Minimizes weight memory bandwidth**
 - Dominant cost at inference (batch = 1 to few)
- **Forward pass only (1 matmul per layer)**
 - N^3 MACs (= $2 \cdot N^3$ FLOPs) for $N \times N$ matmul

Training

- **3 matmuls per layer: 1 forward + 2 backward**
 - $dA = dY \cdot W^T$, $dW = X^T \cdot dY$ (each N^3 MACs)
 - Total: $3 \cdot N^3$ MACs (= $6 \cdot N^3$ FLOPs) per layer
 - Must also store forward activations X for dW
- **No single dominant dataflow**
 - Output- or row-stationary often preferred (Eyeriss)
- **TPU/Trainium: weight-stationary forward, different backward tiling**

Quiz 1



Difficulty: ● Very easy ● Easy ● Moderate ● Hard ● Very hard

Preparation: ● Not at all ● Slightly ● Moderately ● Well ● Very well